BLOCKCHAIN-BASED STUDENT EDUCATION AND CAREER INFORMATION SYSTEM

A PROJECT REPORT

Submitted by

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ABSTRACT

The fast growth of education and certification systems has created a big need for a secure and efficient way to manage student records. The traditional systems used for managing student records have several limitations, including data breaches, unauthorized modifications, certificate forgery, and slow verification processes. These issues reduce trust and efficiency in handling academic and career information. The system is developed as a decentralized application running on the Ethereum blockchain, where student information such as academic qualifications, certifications, skills, internships, and job history can be stored securely. Since blockchain data is immutable, it cannot be altered or tampered with once recorded, ensuring data integrity and trust. The system uses smart contracts to manage and protect student profiles, automating processes such as data storage and verification. For handling certificate documents, the system uses the InterPlanetary File System (IPFS), a decentralized storage network. Instead of storing the actual files on the blockchain, a unique cryptographic hash is stored, which acts as a reference to the original document. This allows users to verify the authenticity of certificates while reducing storage costs. The platform consists of multiple components, including smart contracts, backend services, and a user-friendly interface, all working together to ensure smooth functionality. The system also integrates MetaMask for secure authentication and interaction with the blockchain network. By combining blockchain technology with decentralized storage, the system improves transparency, reduces verification time, prevents data manipulation, and gives students full control over their records.

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LIST OF ABBREVIATIONS

ABBREVIATIONS	FULL FORM
AI	**A**rtificial **I**ntelligence
API	**A**pplication **P**rogramming **I**nterface
BC	**B**lockchain
dApp	**D**ecentralized **A**pplication
DB	**D**atabase
DID	**D**ecentralized **I**dentity
ETH	**E**thereum
EVM	**E**thereum **V**irtual **M**achine
IPFS	**I**nterPlanetary **F**ile **S**ystem
IT	**I**nformation **T**echnology
JS	**J**avaScript
LLM	**L**arge **L**anguage **M**odel
ML	**M**achine **L**earning
NFT	**N**on-**F**ungible **T**oken
NoSQL	**N**ot **O**nly **S**tructured **Q**uery **L**anguage
REST	**R**epresentational **S**tate **T**ransfer
UI	**U**ser **I**nterface
UX	**U**ser **E**xperience
URL	**U**niform **R**esource **L**ocator
HTTP	**H**yperText **T**ransfer **P**rotocol
HTTPS	**H**yperText **T**ransfer **P**rotocol **S**ecure
JSON	**J**avaScript **O**bject **N**otation
RAM	**R**andom **A**ccess **M**emory

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

The education sector has undergone significant transformation due to rapid technological advancements, leading to the generation and storage of vast amounts of student-related data. Educational institutions maintain various records, including academic transcripts, certificates, and job history. Typically, these records are housed in centralized databases managed by universities or specific organizations. While these systems facilitate easy data access and storage, they possess inherent flaws regarding security and transparency.

A critical vulnerability of centralized systems is their susceptibility to hacking and unauthorized data manipulation. Because a single entity often controls the data, the system is susceptible to internal interference. Furthermore, students lack direct autonomy over their own records, requiring them to request school permission to access or share credentials. This dependency often results in administrative delays, particularly during employment applications or transfers to other institutions.

The ease of faking certificates and misrepresenting educational backgrounds presents another major challenge. Modern digital tools have made the forgery of academic documents increasingly simple, making it difficult for employers to verify the authenticity of a candidate's credentials. The traditional verification process, which involves contacting the issuing school directly, is often time-consuming and inefficient. It mitigate these challenges, blockchain technology offers a robust solution. Blockchain serves as a system where data is stored in an immutable format, ensuring that once information is recorded, it remains permanent and trustworthy. Because the technology is decentralized rather than controlled by a single party, users gain greater control over their own data.

1.2 PROBLEM STATEMENT

The education sector is generating a large amount of student data because of the growth of digital technologies and online systems. Institutions maintain many types of records, such as academic transcripts, certificates, skills, and employment history, usually in centralized databases controlled by universities or organizations. These traditional systems make it easy to store and access data, but they suffer from issues like low security, limited transparency, and a high risk of tampering.

In centralized systems, a single authority controls all student information, which creates a single point of failure and makes the data an attractive target for hackers. Records can be changed without permission, and students have very little control over who can view or share their credentials. This leads to delays when applying for jobs or higher studies, because verification often requires direct contact with the issuing institution.

At the same time, fake degrees and forged certificates are becoming more common, since modern tools make it easy to create realistic but fraudulent documents. Employers and universities then struggle to confirm whether a candidate's credentials are genuine, and manual verification takes a lot of time and effort. These problems reduce trust in academic records and slow down recruitment and admission processes.

Blockchain technology offers a promising solution to these challenges by providing a secure, tamper-resistant, and transparent way to store student information. Once data is written to the blockchain, it cannot be altered, which increases trust in the authenticity of academic records. Because blockchain is decentralized, students can have more control over their own data and can share verified credentials directly with employers or institutions.

1.3 GPT4ALL ORCA MINI 3B MODEL

The GPT4ALL model, also known as the ORCA MINI 3B makes the system better by adding automation and natural language interaction. The GPT4ALL is a simple and open-source language model that can be used on your computer, which is great for applications that need to keep things private do not use a lot of resources and can work even without the internet. When we use this model in the system users can work with the platform in a natural and efficient way.

One of the uses of the ORCA MINI 3B model is to provide help to users through chatbots. Students can ask questions about their records, certificates or job opportunities and get answers right away. This means they do not have to look around the system and makes it easier for users who're not good with technology. Employers can also use the chatbot to find information about candidates or check the status of their credentials.

The GPT4ALL model can also be used to analyze resumes and give career advice. By looking at student data such as skills, certificates and experience the system can suggest ways to improve recommend job roles that're a good fit and give personalized career advice. This makes the platform more useful by doing more than storing data and helps users make smart decisions.

1.4 OBJECTIVES OF THE PROJECT

The objectives of the projects are

- To develop a secure blockchain-based platform for managing student career profiles, where each user is identified through a wallet (MetaMask) and profile data is stored on-chain.

- To ensure authenticity and tamper-resistance of certificates by using hashing (Keccak-256) and digital signature verification with Elliptic Curve Digital Signature Algorithm.

- To develop a unified digital profile system where students can add and manage education, skills, and certifications, with verification status stored on-chain.

- To implement a transparent verification mechanism using trusted issuers, oracle-based validation, and admin-controlled verification logic within the smart contract.

- To improve trust and efficiency in recruitment by replacing manual background checks with blockchain-enforced verification and publicly visible credential status.

1.5 SCOPE OF THE PROJECT

This project is about creating a decentralized platform for managing student education and career information using blockchain technology. The goal is to have a way to store, share and verify academic and professional records while keeping the data safe and transparent and giving user control. It involves building a kind of application called a decentralized application or App that lets students make and manage their own digital profiles. These profiles have information like academic qualifications, certifications, skills, internships and job history.

It improves efficiency by reducing the time required for managing and verifying student records. It uses a decentralized storage method called IPFS to store large files such as certificates. Instead of storing the actual documents directly on the blockchain, the system stores a unique cryptographic hash that represents each document. This approach ensures data security, prevents tampering, and significantly reduces storage costs.

It also integrates MetaMask to provide secure user authentication and enable interaction with the blockchain network. Users can log in using their wallet, which ensures safe and authorized access without relying on traditional username and password systems. This makes the platform easy to use for both students and employers, while allowing employers to verify credentials instantly without following time-consuming manual processes.

Blockchain technology plays a key role in ensuring that student education and career information is secure, transparent, and trustworthy. The decentralized nature of the system gives students full control over their data, allowing them to manage and share their records as needed. Overall, the project focuses on using blockchain technology to create a reliable, efficient, and secure solution for managing student education and career information.

CHAPTER 2
LITERATURE SURVEY

[1] Blockchain in Education: Transforming Learning, Credentialing, and Academic Data Management (2024)

W. Suktam et al. (2024), in their paper "Blockchain in Education: Transforming Learning, Credentialing, and Academic Data Management," discussed the transformative impact of blockchain technology in the education sector, particularly in enhancing learning systems and academic data management. The study highlights how blockchain enables decentralized credentialing, allowing educational institutions to issue and verify digital certificates securely without relying on centralized authorities. This approach significantly reduces the risk of data tampering and credential fraud.

[2] Blockchain in Education Systems (2023)

S. Venkatramani and S. Suresh (2023) introduced EduChain, a blockchain-based identity verification model designed to enhance trust and transparency in educational institutions. The study addresses the growing problem of fraudulent academic credentials and the inefficiencies associated with traditional verification systems. EduChain leverages blockchain technology to create a decentralized and tamper-proof system where student records can be securely stored and verified. The proposed model allows institutions to issue digital certificates that are permanently recorded on the blockchain, ensuring their authenticity and preventing unauthorized modifications

[3] Privacy-Protected Blockchain Framework (2021)

R. A. Mishra, A. Kalla, A. Braeken, and M. Liyanage (2021) proposed a privacy-protected blockchain-based architecture for sharing students' credentials securely across institutions. The study focuses on addressing privacy concerns associated with storing sensitive academic data on distributed systems. The authors designed a framework that combines blockchain technology with encryption and access control mechanisms to ensure that only authorized users can access specific information. The system allows students to share their credentials securely while maintaining control over their data. Additionally, the architecture ensures data integrity and prevents unauthorized modifications through blockchain's immutable nature. The study also discusses the use of smart contracts to automate access permissions and verification processes

[4] EduRSS: Educational Record Storage System (2019)

H. Li, D. Han, K.-C. Li, and A. Castiglione (2019) introduced EduRSS, a blockchain-based educational record secure storage and sharing scheme designed to ensure data integrity and security. The study highlights the vulnerabilities of traditional systems, including data tampering and lack of transparency. EduRSS uses blockchain technology to store educational records in a decentralized manner, ensuring that the data cannot be altered or deleted once recorded. The system incorporates cryptographic techniques to enhance security and protect sensitive information.

[5] Blockchain-Based Achievement Record System (2022)

B. Awaji and E. Solaiman (2022) developed a blockchain-based trusted achievement record system aimed at securely storing and managing students' academic accomplishments. The study addresses the limitations of traditional record-keeping systems, which are often vulnerable to data manipulation and unauthorized access. The proposed system uses blockchain technology to create a decentralized and immutable database where student achievements are recorded permanently. This ensures data integrity and prevents any alterations once the information is stored. The authors also emphasize the use of cryptographic techniques to protect sensitive data and maintain privacy. The system allows educational institutions to issue verified records that can be easily accessed and validated by authorized parties.

[6] Certificate Sharing Using Blockchain and NFTs (2023)

P. Khati, A. K. Shrestha, and J. Vassileva (2023) proposed a blockchain-based student certificate sharing system that utilizes Non-Fungible Tokens (NFTs) to enhance the security and ownership of academic credentials. The study focuses on addressing the issue of certificate forgery and the lack of reliable verification mechanisms in traditional systems. By representing certificates as NFTs on a blockchain, the proposed system ensures that each credential is unique, traceable, and tamper-proof. The authors highlight how blockchain technology enables secure storage and easy sharing of certificates between students, institutions, and employers.

[7] Blockchain in Education: Challenges and Opportunities (2021)

J. Park (2021) analyzed the promises and challenges of blockchain technology in education. The study highlights the potential benefits of blockchain, including improved security, transparency, and efficiency. However, it also identifies challenges such as scalability, regulatory issues, and integration complexity. The research emphasizes the need for further studies and development to address these challenges. Overall, the study provides valuable insights into the opportunities and limitations of blockchain in education.

[8] Blockchain-Based Student Identity System (2025)

X. Wang, X. Xu, V. N. Lei, and S. Hao (2025) proposed a conceptual design for a blockchain-based student identity management system (BSIMS), focusing on improving the security and reliability of identity verification in educational institutions. The study highlights the limitations of traditional centralized identity systems, which are prone to data breaches, identity theft, and lack of transparency. To overcome these challenges, the authors designed a decentralized framework where student identities are securely stored on a blockchain network, ensuring immutability and tamper resistance. The proposed system enables secure authentication and verification processes without relying on third-party intermediaries, thereby reducing operational complexity and enhancing trust.

[9] Blockchain-Assisted Self-Sovereign Identity (2025)

W. Chan, K. Gai, J. Yu, and L. Zhu (2025) presented a comprehensive survey on blockchain-assisted self-sovereign identity (SSI) systems in the education sector, focusing on how decentralized technologies can transform traditional identity management approaches. The study highlights that conventional educational systems rely heavily on centralized authorities to store and verify student data, which often leads to issues such as data breaches, lack of transparency, and limited user control.

CHAPTER 3

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

The existing system for managing student education and career information is mainly based on centralized databases controlled by schools, universities, and certification authorities. These systems are used to store and manage student records such as academic transcripts, certificates, grades, attendance, and other related information. Although centralized systems are simple and easy to implement, they have several limitations in terms of security, transparency, and efficiency. In such systems, all student data is stored in a single database managed by an institution, and access is restricted to administrators. This means that students do not have direct control over their own records and must depend on the institution whenever they need to access or share their credentials, which can lead to delays and inefficiencies.

Another major issue in the existing system is the lack of transparency. Since only authorized personnel can update or modify records, it is difficult to track how data is handled. This increases the risk of unauthorized changes or errors in the data. Mistakes in data entry or updates may go unnoticed, which can affect the accuracy and reliability of student records. The verification process is also time-consuming, as employers or institutions must manually contact the issuing authority to confirm the authenticity of certificates. This process can take several days or even weeks, causing delays in recruitment and admission procedures.

In addition, centralized systems are highly vulnerable to security threats. Since all data is stored in one location, it becomes an easy target for cyberattacks, and any data breach can result in the loss or exposure of sensitive information. There is also a risk of unauthorized modification or deletion of records. Certificate forgery is another significant issue, as fake certificates can be created or altered easily, and due to slow verification processes, it is difficult to detect such fraud in time. Furthermore, centralized systems face scalability issues, as they struggle to efficiently handle large volumes of data and are difficult to integrate with other systems due to lack of standardization. Overall, these limitations highlight the need for a more secure, transparent, and efficient system.

3.2 PROPOSED SYSTEM

The proposed system, known as the Blockchain-Based Student Education and Career Information System, is designed to overcome the limitations of traditional centralized systems by providing a secure, transparent, and decentralized platform for managing student data. The system utilizes advanced technologies such as blockchain, smart contracts, and decentralized storage to ensure reliability and efficiency. In this system, student records are stored on the Ethereum blockchain, which ensures that once the data is recorded, it cannot be altered or deleted without authorization. Each student is assigned a unique wallet address that acts as their digital identity and gives them full control over their personal data.

The system follows a three-layer architecture consisting of the frontend, backend, and blockchain layers. The frontend provides an interactive and user-friendly interface for students and employers, while the backend manages communication between the frontend and blockchain. The core functionality is handled by smart contracts, which are self-executing programs deployed on the blockchain. These smart contracts automate operations such as profile creation, certificate storage, and credential verification, eliminating the need for intermediaries and ensuring secure and transparent transactions.

Blockchain efficiently store large files such as certificates, the system uses the InterPlanetary File System (IPFS), a decentralized storage network. Instead of storing files directly on the blockchain, the system stores them in IPFS and records their unique hash values on the blockchain. This approach ensures data integrity while reducing storage costs. The system also integrates MetaMask for secure authentication, allowing users to log in using their blockchain wallet and manage their data safely.

The proposed system provides several advantages over traditional systems, including enhanced security, data integrity, and transparency. It enables instant verification of credentials, reduces the risk of fraud, and gives students full ownership of their data. Additionally, the system is scalable, cost-efficient, and easily accessible from anywhere. By combining blockchain technology with modern web development, the proposed system offers a reliable and efficient solution for managing student education and career information.

CHAPTER 4

SYSTEM DESIGN

4.1 GENERAL

The software component of the Blockchain-Based Student Education and Career Information System plays a crucial role in building, integrating, and operating the entire application. The system is designed as a full-stack decentralized application (dApp), combining frontend, backend, and blockchain technologies to ensure security, transparency, and efficient performance. The frontend of the system is developed using modern web technologies such as React.js and TypeScript, which provide a responsive and user-friendly interface. This enables students and employers to easily perform tasks such as creating profiles, uploading certificates, and verifying credentials.

The backend is developed using Node.js and Express.js, which handle server-side operations and ensure smooth communication between different components of the system. It provides APIs for managing user requests, processing data, and interacting with blockchain and storage systems. The core functionality of the system is powered by the Ethereum blockchain, where smart contracts written in Solidity are deployed using tools like Hardhat. These smart contracts automate operations such as storing student records, managing certifications, and verifying credentials in a secure and transparent manner.

Blockchain efficiently store large files such as certificates, the system uses the InterPlanetary File System (IPFS), a decentralized storage solution that generates unique hash values for each file. These hash values are recorded on the blockchain to ensure data integrity. Additionally, MongoDB is used to store application-related data such as user details and logs. By integrating these technologies, the system provides a secure, scalable, and efficient solution for managing student education and career information.

4.2 HARDWARE REQUIREMENTS

The hardware requirements for the proposed system are designed to ensure smooth development and efficient usage. Since the system is web-based and uses blockchain technology, it does not require highly specialized hardware, but a standard computing environment is necessary for reliable performance. For development purposes, a system with at least an Intel Core i3 processor or equivalent is required, while a higher configuration such as Intel i5 or AMD Ryzen 5 is recommended for better performance. The system should have a minimum of 4 GB RAM, although 8 GB RAM is preferred to support multitasking and handle blockchain-related operations efficiently.

A storage capacity of at least 256 GB, preferably a solid-state drive (SSD), is required to store development tools, libraries, and project files. An SSD improves system speed and responsiveness. A stable internet connection is essential, as the system interacts with blockchain networks, decentralized storage (IPFS), and online development tools. Continuous connectivity ensures smooth communication between all components.

The system can be accessed using devices such as desktops, laptops, or mobile devices with modern web browsers. Browsers like Google Chrome or Mozilla Firefox are recommended, as they support blockchain wallets such as MetaMask. Basic input and output devices like a keyboard, mouse, and monitor are required for user interaction. Advanced hardware such as GPUs is not mandatory but can enhance performance if additional features like AI integration are implemented. Overall, the hardware requirements are minimal and cost-effective, making the system accessible to a wide range of users.

4.3 SOFTWARE REQUIREMENTS

The proposed Blockchain-Based Student Education and Career Information System is developed using a combination of modern software technologies that work together to ensure security, efficiency, and scalability. Each component plays a specific role in the overall functioning of the system, including frontend, backend, blockchain, storage, and authentication technologies.

4.3.1 FRONTEND TECHNOLOGIES

The frontend of the Blockchain-Based Student Education and Career Information System is developed using React.js and TypeScript, which provide a modern, responsive, and user-friendly interface. React.js is a widely used JavaScript library that enables the development of dynamic and interactive web applications through reusable components. This improves performance and makes the application easier to maintain. TypeScript, which is an extension of JavaScript, adds type safety and helps detect errors during the development stage, resulting in more reliable and maintainable code.

4.3.2 BACKEND TECHNOLOGIES

The backend of the Blockchain-Based Student Education and Career Information System is developed using Node.js and Express.js, which handle all server-side operations and ensure smooth communication between different components of the system. Node.js provides a fast and scalable runtime environment that can efficiently manage multiple user requests simultaneously. It is well-suited for real-time applications and helps improve the overall performance of the system. Express.js, a lightweight web framework for Node.js, simplifies the development of APIs and routing, making it easier to build and manage server-side functionalities.

4.3.3 BLOCKCHAIN PLATFORM

The Ethereum blockchain serves as the core platform for the Blockchain-Based Student Education and Career Information System, providing a secure and decentralized environment for storing and managing critical data. Ethereum is a widely used blockchain network that supports smart contracts and decentralized applications (dApps). It ensures that all data stored on the network is immutable, meaning it cannot be altered or deleted once it has been recorded. This feature plays a crucial role in maintaining the integrity and authenticity of student records.

4.3.4 SMART CONTRACTS

The Smart Contracts component is a fundamental part of the Blockchain-Based Student Education and Career Information System, enabling automation and secure execution of system operations. Smart contracts are self-executing programs deployed on the blockchain that automatically perform predefined actions when specific conditions are met. In this system, smart contracts are written using Solidity, a programming language specifically designed for developing applications on the Ethereum blockchain.

4.3.5 DECENTRALIZED STORAGE

The Decentralized Storage component of the system is implemented using the InterPlanetary File System (IPFS), which provides a secure and efficient way to store large files such as student certificates and documents. Instead of storing files directly on the blockchain, which is expensive and inefficient, IPFS stores files in a distributed network of nodes. This approach reduces storage costs and improves system performance while maintaining high security.

4.3.6 DATABASE

The system uses MongoDB to store information like user details, logs and metadata. It is a fast database that works well with the blockchain. It helps manage information that's not critical like user details and logs.

4.3.7 AUTHENTICATION

The system uses MetaMask, a wallet to let users log in and interact with the blockchain. It is a way for users to access their information. Each user has a wallet address, which means that they have secure access to their information.

4.3.8 DEVELOPMENT TOOLS

The development of the Blockchain-Based Student Education and Career Information System relies on a set of essential tools that support efficient coding, version control, and project management. These tools help developers build, test, and maintain the system in an organized and effective manner. One of the primary tools used is Visual Studio Code (VS Code), a lightweight and powerful code editor that supports multiple programming languages. It provides features such as syntax highlighting, debugging, extensions, and integrated terminal support, which improve developer productivity and make coding easier.

4.3.9 WEB BROWSER

This Blockchain need a web browser like Google Chrome to run. It supports MetaMask. Works well, with the web technologies used in the system.

CHAPTER 5
SYSTEM IMPLEMENTATION

5.1 PROJECT DESCRIPTION

The proposed project, "Blockchain-Based Student Education and Career Information System," is designed to provide a secure and reliable platform for managing student academic and professional records. In traditional systems, educational institutions store student data in centralized databases, which are vulnerable to data loss, unauthorized access, and manipulation. These systems lack transparency and do not provide full control to students over their own records. As a result, verifying academic credentials becomes a time-consuming and inefficient process.

Blockchain address these challenges, the proposed system uses blockchain technology to ensure secure, transparent, and tamper-proof data storage. The system is developed as a decentralized application (dApp) using the Ethereum blockchain, where all critical data is stored in an immutable format. This means that once the data is recorded, it cannot be altered or deleted without proper authorization, thereby ensuring data integrity and trust.

In this project, students can create their own digital profiles and add important information such as academic qualifications, certificates, skills, internships, and job experiences. Smart contracts are used to manage and store this data securely, enabling automated and transparent operations without the need for intermediaries. These contracts ensure that all actions performed within the system are recorded and verified.

Blockchain handle large files such as certificates, the system integrates the InterPlanetary File System (IPFS), which provides decentralized storage. Files are stored in IPFS, and a unique cryptographic hash is generated and stored on the blockchain. This allows users to verify the authenticity of documents efficiently.

The system also uses MetaMask for secure authentication, enabling users to access their accounts and perform transactions safely. Additionally, employers and institutions can instantly verify student credentials without manual processes, reducing time and effort. Overall, the system improves security, transparency, and efficiency while giving students full control over their records.

5.1.1 METHODOLOGY

The methodology of the proposed Blockchain-Based Student Education and Career Information System follows a structured approach to design and develop a secure, decentralized, and efficient platform for managing student records. The system is implemented as a decentralized application (dApp) that integrates frontend, backend, blockchain, and decentralized storage technologies to ensure data integrity, transparency, and reliability.

Initially, the system is designed with a three-layer architecture consisting of the frontend, backend, and blockchain layers. The frontend provides a user-friendly interface where students and administrators can perform operations such as creating profiles, uploading certificates, and viewing records. The backend handles the application logic and acts as a bridge between the frontend and blockchain, managing data flow and processing user requests.

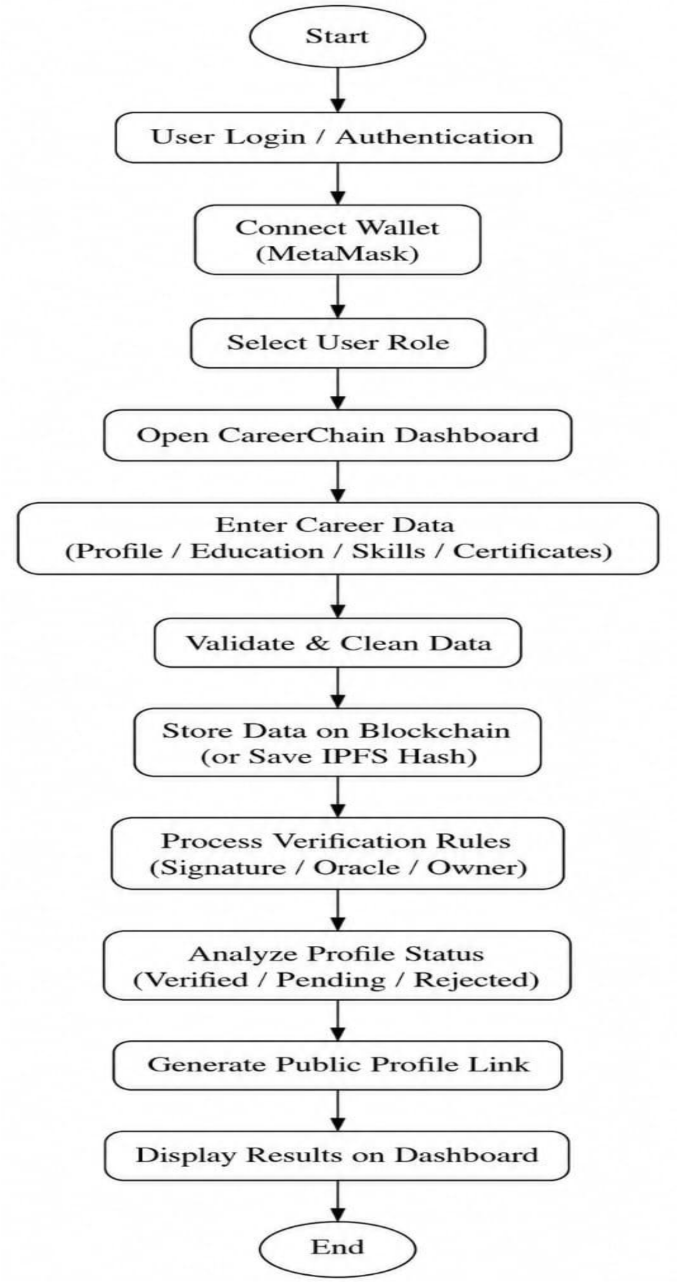


Figure 5.1.1 Flow Diagram

The Figure 5.1.1 represents user logs in, connects wallet, and accesses the Career Chain dashboard. Career details are entered, validated, and securely stored on blockchain/IPFS with verification. Profile is analyzed, public link generated, and results displayed.

5.2 MODULE DESCRIPTION

5.2.1 USER AUTHENTICATION MODULE

The User Authentication Module plays an important role in maintaining the security of the Blockchain-Based Student Education and Career Information System. Unlike traditional systems that use usernames and passwords, this system uses a more secure method based on blockchain wallets such as MetaMask. Each user is provided with a unique wallet address, which acts as their digital identity within the system. When a user wants to access the platform, they must connect their MetaMask wallet and approve the login request. This process verifies the user's identity using cryptographic methods, making it more secure and reducing the risk of hacking or data breaches.

Once the user is authenticated, they are allowed to perform actions such as creating profiles, uploading certificates, and viewing their records. The module ensures that only the owner of the wallet can access and modify their data, providing full control and privacy to the user. Since the system does not rely on centralized storage of login credentials, it eliminates common issues such as password theft and unauthorized access.

5.2.2 STUDENT PROFILE MANAGEMENT MODULE

The Student Profile Management Module is responsible for creating and maintaining the digital profiles of students within the system. This module allows users to store and manage all important academic and professional details in one place. Students can create their profiles and add information such as educational qualifications, certifications, skills, internships, and job experiences. The interface is designed to be simple and user-friendly, enabling students to easily add, update, and view their information whenever required.

All the data entered by students is stored securely using blockchain technology, ensuring that it remains tamper-proof and cannot be altered without authorization. Each student profile is linked to a unique blockchain wallet address, which acts as their identity in the system. This ensures that only the respective student can manage their profile, providing complete ownership and control over their data.

The module also validates the information before storing it, ensuring accuracy and consistency. It allows real-time updates so that students can keep their profiles up to date with their latest achievements. Authorized users such as employers or institutions can view the profile for verification purposes. Overall, the Student Profile Management Module provides a secure, transparent, and efficient way for students to manage their education and career information.

5.2.3 CERTIFICATION MANAGEMENT MODULE

The Certification Management Module is responsible for handling the upload, storage, and verification of student certificates and related documents within the system. This module allows students to securely upload their academic certificates, training records, and other credentials in digital format. Instead of storing these files directly on the blockchain, which can be costly and inefficient, the system uses a decentralized storage solution called the Inter Planetary File System (IPFS).

When a student uploads a certificate, the file is stored in IPFS, and a unique cryptographic hash is generated for that document. This hash value is then recorded on the blockchain using smart contracts. The hash acts as a digital fingerprint of the file, ensuring that the document remains authentic and unchanged.

5.2.4 SMART CONTRACT MODULE

The Smart Contract Module is a core component of the Blockchain-Based Student Education and Career Information System, responsible for automating and securing all major operations within the platform. Smart contracts are self-executing programs deployed on the blockchain that run automatically when predefined conditions are met. In this system, smart contracts are used to manage student profiles, store certification data, and handle the verification process without the need for intermediaries.

When a student creates a profile or uploads a certificate, the smart contract records the necessary information on the blockchain. These contracts ensure that all data is stored in a structured and secure manner, and once recorded, it cannot be altered or deleted. This guarantees data integrity and prevents unauthorized modifications. The smart contract also controls access permissions, ensuring that only authorized users can update or view specific data.

5.2.5 SCORING AND FEEDBACK MODULE

The Scoring and Feedback Module is designed to evaluate and present meaningful insights based on the data available in the Blockchain-Based Student Education and Career Information System. This module helps in analyzing student performance by considering various factors such as certifications, skills, academic achievements, and overall profile completeness. It provides a structured way to assess a student's profile, making it easier for employers and institutions to understand their qualifications and capabilities.

Additionally, the module supports transparency by presenting evaluation results in a clear and easy-to-understand format. It may also allow comparisons between students or track individual progress over time. These features help both students and administrators gain valuable insights into performance trends. Overall, the Scoring and Feedback Module improves decision-making, encourages skill development, and enhances the overall effectiveness of the system.

5.2.6 DATABASE MANAGEMENT MODULE

The Database Management Module is responsible for storing, organizing, and managing all data within the Blockchain-Based Student Education and Career Information System. It works alongside the blockchain to ensure efficient data handling by using a hybrid approach. While critical data such as certificate hashes and verification records are stored securely on the blockchain, non-critical data like user profiles, login details, and system logs are managed using a database such as MongoDB. This combination improves system performance while maintaining high security.

The module ensures that data is structured properly, allowing fast retrieval, updating, and processing of information. It supports real-time data access, enabling smooth communication between different components of the system such as the frontend, backend, and blockchain layers. This ensures that users can quickly access their information without delays.

5.2.7 ADMIN DASHBOARD MODULE

The Admin Dashboard Module provides a centralized interface for administrators and institutions to manage and monitor the Blockchain-Based Student Education and Career Information System. It allows authorized users to oversee student data, verify credentials, and ensure that the system operates smoothly. Through the dashboard, administrators can view student profiles, certification details, and verification status, enabling them to maintain the authenticity and reliability of the data stored in the system.

CHAPTER 6

RESULTS AND DISCUSSION

6.1 RESULTS

The proposed Blockchain-Based Student Education and Career Information System was successfully implemented as a decentralized application. The system demonstrates how blockchain technology can be effectively used to securely store, manage, and verify student academic and professional records. One of the key outcomes of the system is the secure storage of data. By using blockchain technology, important information such as certifications and verification records is stored in an immutable format, ensuring that once t4he data is recorded, it cannot be altered or tampered with. This enhances data integrity and builds trust among users.

The final output of the project is a fully functional blockchain-based web application where students can create a career profile using a wallet metamask, add their education, skills, and certifications, and have these records securely stored and managed on-chain. Certificates can be verified using cryptographic methods such as hashing with Keccak-256 and signature validation using Elliptic Curve Digital Signature Algorithm, ensuring authenticity and preventing tampering. The system generates a public profile link that can be shared with employers, allowing them to instantly view and verify credential status verified, pending, or revoked without login, making the entire process transparent, secure, and efficient compared to traditional resume verification methods.

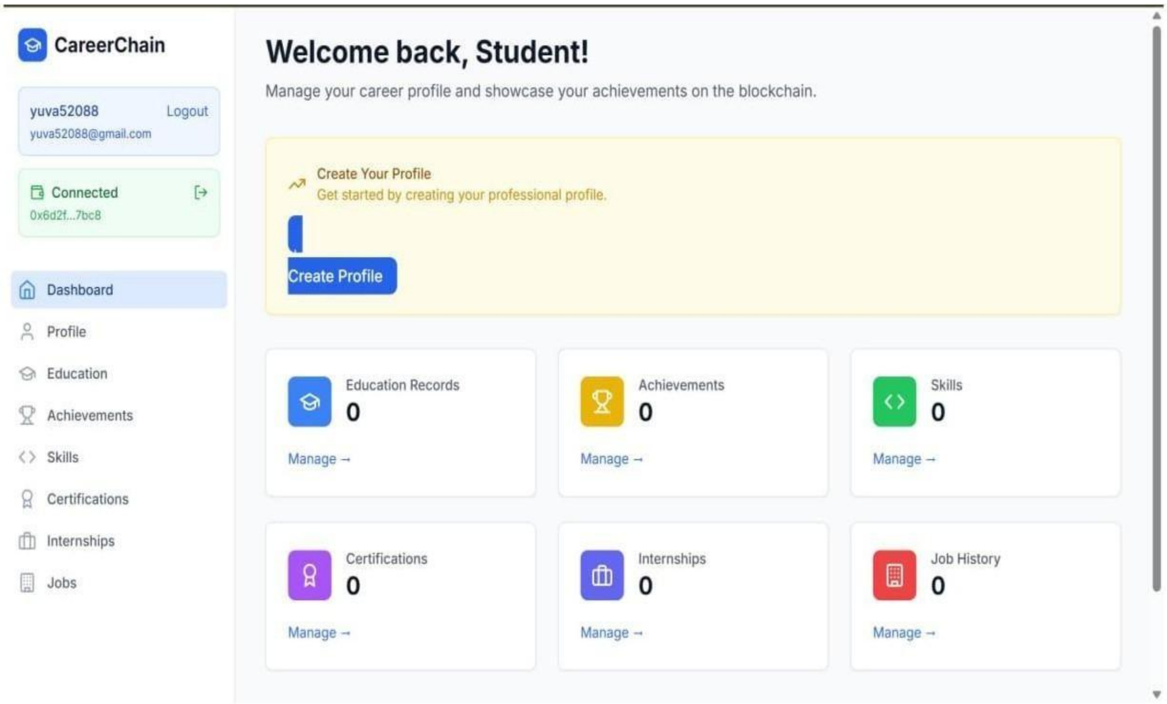


Figure 6.1.1 Web Dashboard

The figure 6.1.1 represents the dashboard provides a centralized platform for students to manage and showcase their career profile on the blockchain. Users can add and track education records, achievements, skills, certifications, internships, and job history in one place. It ensures a secure, transparent, and organized representation of professional growth and accomplishments.

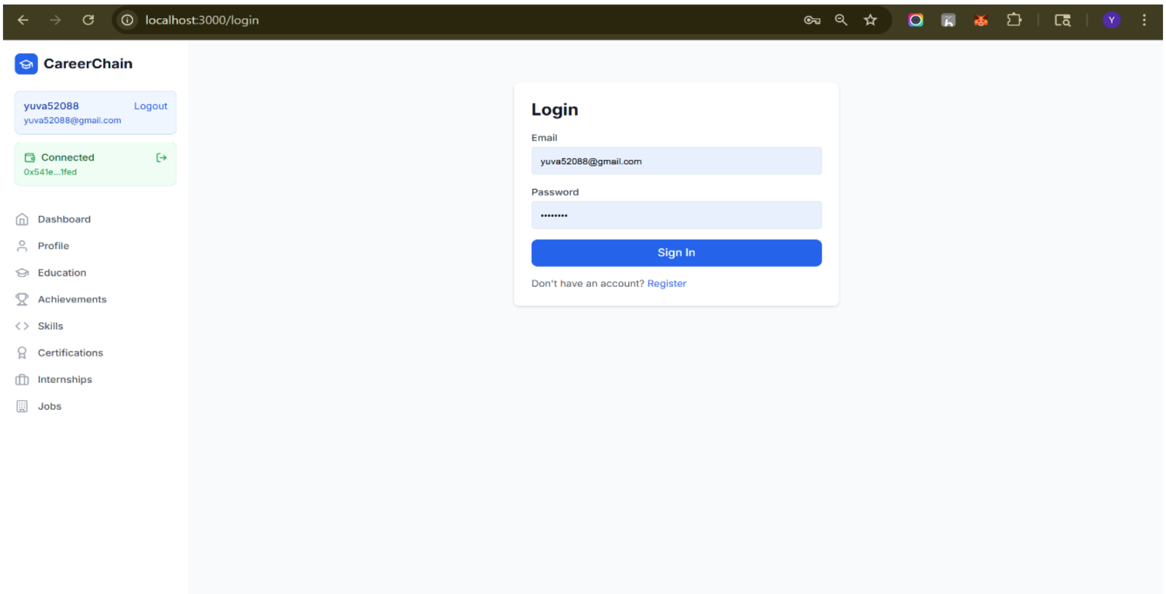


Figure 6.1.2 Login page

The Figure 6.1.2 represents the login page contains input fields for email and password, where registered users can enter their credentials to access their accounts. Once the correct details are entered, the user can click the "Sign In" button to log in to the system. The system validates the credentials through the backend and grants access only to authorized users, ensuring security.

CHAPTER 7
CONCLUSION AND FUTURE EXHANCEMENT

7.1 CONCLUSION

This system thoroughly tested the core functionalities of the blockchain-based career profile system to ensure reliability and correctness. Using a Hardhat test suite, we validated profile creation and updates, prevention of duplicate profiles, addition of education, skills, and certifications, and proper management of trusted issuers and oracles, also tested certificate verification workflows, including valid and invalid signature checks using Elliptic Curve Digital Signature Algorithm, oracle-based verification authorization, and revocation handling. These tests confirm that the smart contract enforces all trust rules correctly.

The system achieves high accuracy in verification, as it relies on deterministic cryptographic methods rather than probabilistic or manual processes. Certificate data is hashed using Keccak-256, ensuring that even a minor change invalidates the signature. Since only the holder of the issuer's private key can generate a valid signature, the verification accuracy is effectively near 100%, assuming trusted key management. This eliminates human errors and significantly reduces the chances of fraud or false verification.

This system successfully demonstrates how blockchain technology can transform traditional resume systems into secure, transparent, and verifiable digital profiles. By combining smart contracts, cryptographic verification, and public accessibility, the system improves trust and efficiency in credential validation. The implementation proves that verification can be automated, tamper-resistant, and instantly accessible, making it a strong foundation for future enhancements such as expanded record types, improved privacy controls, and real-world deployment on public networks.

7.2 FUTURE ENHANCEMENT

The proposed Blockchain-Based Student Education and Career Information System provides a strong foundation for secure and decentralized record management, but there are several opportunities for further improvement and expansion. One important enhancement is the integration of multiple blockchain networks, which would improve scalability and allow the system to handle a larger number of users across different platforms.

The incorporation of Artificial Intelligence (AI) and Machine Learning (ML) can further enhance the system by providing personalized career recommendations, skill analysis, and job matching based on student profiles. These technologies can also assist institutions and employers in making better decisions through advanced data analysis.

REFERENCES

[1] W. Chan, K. Gai, J. Yu, and L. Zhu, "Blockchain-assisted self-sovereign identities in education: A survey," *Blockchains*, vol. 3, no. 1, p. 3, Feb. 2025.

[2] X. Wang, X. Xu, V. N. Lei, and S. Hao, "Conceptual design of a blockchain-based student identity management system (BSIMS)," *Discover Computing*, vol. 28, no. 42, Apr. 2025.

[3] S. Venkatramani and S. Suresh, "EduChain: A blockchain-based identity verification model for educational institutions," *in Proceedings of the International Conference on Data Science Applications*, 2023.

[4] P. Khati, A. K. Shrestha, and J. Vassileva, “Student certificate sharing system using blockchain and NFTs,” 2023.

[5] B. Awaji and E. Solaiman, “Design, implementation, and evaluation of a blockchain-based trusted achievement record system for students,” 2022.

[6] R. A. Mishra, A. Kalla, A. Braeken, and M. Liyanage, “Privacy-protected blockchain-based architecture for sharing students’ credentials,” *Information Processing and Management*, vol. 58, no. 3, p. 102512, May 2021.

[7] H. Li, D. Han, K.-C. Li, and A. Castiglione, “EduRSS: A blockchain-based educational records secure storage and sharing scheme,” *IEEE Access*, vol. 7, 2019.

[8] A. de Alwis, A. Shrestha, and T. Sarker, “Exploring governance for accreditation in the education sector using blockchain technology: A systematic literature review,” *Discover Education*, vol. 4, no. 57, Mar. 2025.

[9] W. Suktam, S. Lapchit, J. Supsin, and S. Sonwa, “Blockchain in education: Transforming learning, credentialing, and academic data management,” *Journal of Education and Learning Review*, vol. 1, no. 6, Nov.–Dec. 2024.

[10] J. Park, “Promises and challenges of blockchain in education,” *Smart Learning Environments*, vol. 8, no. 1, p. 33, Dec. 2021.

[11] V. Bhosale and P. Jamsandekar, “Blockchain for student information systems: The need for technological adoption,” *Journal of Pharmaceutical Negative Results*, vol. 13, no. 5, Oct. 2022

APPENDIX -1

```
import React from 'react';
import { BrowserRouter as Router, Routes, Route } from 'react-router-dom';
import { Toaster } from 'react-hot-toast';
import Layout from './components/Layout';
import Dashboard from './pages/Dashboard';
import Profile from './pages/Profile';
import Education from './pages/Education';
import Achievements from './pages/Achievements';
import Skills from './pages/Skills';
import Certifications from './pages/Certifications';
import Internships from './pages/Internships';
import Jobs from './pages/Jobs';
import { Web3Provider } from './contexts/Web3Context';
import { AuthProvider } from './contexts/AuthContext';
import Login from './pages/Login';
import Register from './pages/Register';
import ContractTest from './pages/ContractTest';
import PublicProfile from './pages/PublicProfile';

function App() {
 return (
  <Web3Provider>
   <AuthProvider>
    <Router>
    <div className="min-h-screen bg-gray-50">
     <Layout>
      <Routes>
       <Route path="/" element={<Dashboard />} />
       <Route path="/login" element={<Login />} />
       <Route path="/register" element={<Register />} />
       <Route path="/profile" element={<Profile />} />
       <Route path="/education" element={<Education />} />
       <Route path="/achievements" element={<Achievements />} />
       <Route path="/skills" element={<Skills />} />
       <Route path="/certifications" element={<Certifications />} />
       <Route path="/internships" element={<Internships />} />
       <Route path="/jobs" element={<Jobs />} />
       <Route path="/contract-test" element={<ContractTest />} />
       <Route path="/profile/public/:address" element={<PublicProfile />} />
```

```
            <Route path="/share/:address" element={<PublicProfile />} />
          </Routes>
        </Layout>
        <Toaster
          position="top-right"
          toastOptions={{
            duration: 4000,
            style: {
              background: '#363636',
              color: '#fff',
            },
          }}
        />
      </div>
      </Router>
    </AuthProvider>
  </Web3Provider>
 );
}

export default App;
```

```
import React, { useState, useEffect } from 'react';
import api from '../services/api';

interface CertificateVerificationProps {
  studentAddress: string;
  certificateIndex: number;
  certificate: {
    name: string;
    issuer: string;
    credentialId: string;
    issueDate: number;
    expiryDate: number;
    description: string;
    certificateHash: string;
    isVerified: boolean;
  };
  onVerificationUpdate: () => void;
}
```

```
interface VerificationStats {
  verified: number;
  revoked: number;
  totalIssuers: number;
}

interface TrustedIssuer {
  name: string;
  address: string;
}

const CertificateVerification: React.FC<CertificateVerificationProps> = ({
  studentAddress,
  certificateIndex,
  certificate,
  onVerificationUpdate
}) => {
  const [isVerifying, setIsVerifying] = useState(false);
    const [verificationMethod, setVerificationMethod] = useState<'signature' |
'oracle'>('signature');
  const [signature, setSignature] = useState('');
    const [verificationStats, setVerificationStats] = useState<VerificationStats |
null>(null);
  const [trustedIssuers, setTrustedIssuers] = useState<TrustedIssuer[]>([]);
  const [showAdminPanel, setShowAdminPanel] = useState(false);
  const [newIssuer, setNewIssuer] = useState({ name: '', address: '' });
  const [newOracle, setNewOracle] = useState('');

  useEffect(() => {
   loadVerificationData();
  }, []);

  const loadVerificationData = async () => {
   try {
    const [statsResponse, issuersResponse] = await Promise.all([
     api.get('/verification/stats'),
     api.get('/verification/issuers')
    ]);

    setVerificationStats(statsResponse.data.stats);
    setTrustedIssuers(issuersResponse.data.issuers);
   } catch (error) {
    console.error('Failed to load verification data:', error);
```

```
    }
  };

  const handleSignatureVerification = async () => {
    if (!signature.trim()) {
      alert('Please provide a signature');
      return;
    }

    setIsVerifying(true);
    try {
      await api.post('/verification/signature', {
        studentAddress,
        certificateIndex,
        signature
      });

      alert('Certificate verified successfully with signature!');
      onVerificationUpdate();
    } catch (error: any) {
      console.error('Signature verification failed:', error);
      alert(`Verification failed: ${error.response?.data?.message || error.message}`);
    } finally {
      setIsVerifying(false);
    }
  };

  const handleOracleVerification = async () => {
    setIsVerifying(true);
    try {
      await api.post('/verification/oracle', {
        studentAddress,
        certificateIndex,
        isValid: true
      });

      alert('Certificate verification submitted via oracle!');
      onVerificationUpdate();
    } catch (error: any) {
      console.error('Oracle verification failed:', error);
      alert(`Oracle verification failed: ${error.response?.data?.message || error.message}`);
    } finally {
```

```
    setIsVerifying(false);
  }
};

const handleRevokeCertificate = async () => {
  if (!window.confirm('Are you sure you want to revoke this certificate?')) {
    return;
  }

  setIsVerifying(true);
  try {
    await api.post('/verification/revoke', {
      studentAddress,
      certificateIndex
    });

    alert('Certificate revoked successfully!');
    onVerificationUpdate();
  } catch (error: any) {
    console.error('Certificate revocation failed:', error);
    alert(`Revocation failed: ${error.response?.data?.message || error.message}`);
  } finally {
    setIsVerifying(false);
  }
};

const handleAddTrustedIssuer = async () => {
  if (!newIssuer.name.trim() || !newIssuer.address.trim()) {
    alert('Please provide both issuer name and address');
    return;
  }

  try {
    await api.post('/verification/issuers', newIssuer);
    alert('Trusted issuer added successfully!');
    setNewIssuer({ name: '', address: '' });
    loadVerificationData();
  } catch (error: any) {
    console.error('Failed to add trusted issuer:', error);
    alert(`Failed to add issuer: ${error.response?.data?.message || error.message}`);
  }
};
```

```
const handleAddOracle = async () => {
  if (!newOracle.trim()) {
    alert('Please provide oracle address');
    return;
  }

  try {
    await api.post('/verification/oracles', { oracleAddress: newOracle });
    alert('Verification oracle added successfully!');
    setNewOracle('');
    loadVerificationData();
  } catch (error: any) {
    console.error('Failed to add oracle:', error);
    alert(`Failed to add oracle: ${error.response?.data?.message || error.message}`);
  }
};

const generateSignatureData = async () => {
  try {
    const response = await api.post('/verification/generate-signature-data', {
      studentAddress,
      certificateIndex
    });

    const { messageHash, certificateData } = response.data;

    // Display signature data for issuer to sign
    const signatureData = {
      messageHash,
      certificateData,
      instructions: 'Please sign this message hash with your private key to verify the
certificate.'
    };

    alert(`Signature Data:\n${JSON.stringify(signatureData, null, 2)}`);
  } catch (error: any) {
    console.error('Failed to generate signature data:', error);
    alert(`Failed to generate signature data: ${error.response?.data?.message ||
error.message}`);
  }
};

const isIssuerTrusted = trustedIssuers.some(issuer => issuer.name ===
```

```
certificate.issuer);

  return (
    <div className="bg-white rounded-lg shadow-md p-6">
      <div className="flex justify-between items-center mb-4">
                <h3 className="text-lg font-semibold text-gray-800">Certificate
Verification</h3>
        <div className="flex items-center space-x-2">
          <span className={`px-2 py-1 rounded-full text-xs font-medium ${
            certificate.isVerified
              ? 'bg-green-100 text-green-800'
              : 'bg-yellow-100 text-yellow-800'
          }`}>
            {certificate.isVerified ? 'Verified' : 'Unverified'}
          </span>
          {isIssuerTrusted && (
              <span className="px-2 py-1 rounded-full text-xs font-medium bg-blue-100
text-blue-800">
              Trusted Issuer
            </span>
          )}
        </div>
      </div>

      {/* Certificate Details */}
      <div className="mb-6 p-4 bg-gray-50 rounded-lg">
        <h4 className="font-medium text-gray-700 mb-2">Certificate Details</h4>
        <div className="grid grid-cols-2 gap-4 text-sm">
          <div>
            <span className="font-medium">Name:</span> {certificate.name}
          </div>
          <div>
            <span className="font-medium">Issuer:</span> {certificate.issuer}
          </div>
          <div>
                        <span className="font-medium">Credential ID:</span>
{certificate.credentialId}
          </div>
          <div>
                    <span className="font-medium">Issue Date:</span> {new
Date(certificate.issueDate * 1000).toLocaleDateString()}
          </div>
          <div>
```

```
                    <span className="font-medium">Expiry Date:</span> {new
Date(certificate.expiryDate * 1000).toLocaleDateString()}
          </div>
          <div>
            <span className="font-medium">Hash:</span> {certificate.certificateHash}
          </div>
        </div>
      </div>

      {/* Verification Methods */}
      {!certificate.isVerified && (
        <div className="mb-6">
          <h4 className="font-medium text-gray-700 mb-3">Verification Methods</h4>

          {/* Method Selection */}
          <div className="flex space-x-4 mb-4">
            <label className="flex items-center">
              <input
                type="radio"
                value="signature"
                checked={verificationMethod === 'signature'}
                    onChange={(e) => setVerificationMethod(e.target.value as 'signature' |
'oracle')}
                className="mr-2"
              />
              Cryptographic Signature
            </label>
            <label className="flex items-center">
              <input
                type="radio"
                value="oracle"
                checked={verificationMethod === 'oracle'}
                    onChange={(e) => setVerificationMethod(e.target.value as 'signature' |
'oracle')}
                className="mr-2"
              />
              Oracle Verification
            </label>
          </div>

          {/* Signature Verification */}
          {verificationMethod === 'signature' && (
            <div className="space-y-4">
```

```
          <div>
            <label className="block text-sm font-medium text-gray-700 mb-2">
              Signature (Hex)
            </label>
            <textarea
              value={signature}
              onChange={(e) => setSignature(e.target.value)}
              placeholder="Enter the signature from the issuer..."
              className="w-full p-3 border border-gray-300 rounded-md focus:ring-2
focus:ring-blue-500 focus:border-transparent"
              rows={3}
            />
          </div>
          <div className="flex space-x-2">
            <button
              onClick={generateSignatureData}
              className="px-4 py-2 bg-gray-500 text-white rounded-md hover:bg-gray-
600 transition-colors"
            >
              Generate Signature Data
            </button>
            <button
              onClick={handleSignatureVerification}
              disabled={isVerifying}
              className="px-4 py-2 bg-blue-500 text-white rounded-md hover:bg-blue-
600 transition-colors disabled:opacity-50"
            >
              {isVerifying ? 'Verifying...' : 'Verify with Signature'}
            </button>
          </div>
        </div>
      )}

      {/* Oracle Verification */}
      {verificationMethod === 'oracle' && (
        <div>
          <p className="text-sm text-gray-600 mb-4">
            Submit certificate for verification by trusted oracles.
          </p>
          <button
            onClick={handleOracleVerification}
            disabled={isVerifying}
            className="px-4 py-2 bg-green-500 text-white rounded-md hover:bg-green-
```

```
600 transition-colors disabled:opacity-50"
              >
                {isVerifying ? 'Submitting...' : 'Submit for Oracle Verification'}
              </button>
            </div>
          )}
        </div>
      )}

      {/* Revoke Certificate */}
      {certificate.isVerified && (
        <div className="mb-6">
          <button
            onClick={handleRevokeCertificate}
            disabled={isVerifying}
              className="px-4 py-2 bg-red-500 text-white rounded-md hover:bg-red-600
transition-colors disabled:opacity-50"
          >
            {isVerifying ? 'Revoking...' : 'Revoke Certificate'}
          </button>
        </div>
      )}

      {/* Admin Panel */}
      <div className="border-t pt-4">
        <button
          onClick={() => setShowAdminPanel(!showAdminPanel)}
          className="text-sm text-blue-600 hover:text-blue-800"
        >
          {showAdminPanel ? 'Hide' : 'Show'} Admin Panel
        </button>

        {showAdminPanel && (
          <div className="mt-4 space-y-4">
            {/* Verification Stats */}
            {verificationStats && (
              <div className="p-4 bg-blue-50 rounded-lg">
                <h5 className="font-medium text-blue-800 mb-2">Verification
Statistics</h5>
                <div className="grid grid-cols-3 gap-4 text-sm">
                  <div>
                    <span className="font-medium">Verified:</span>
{verificationStats.verified}
```

```
              </div>
              <div>
                                   <span    className="font-medium">Revoked:</span>
{verificationStats.revoked}
              </div>
              <div>
                              <span   className="font-medium">Trusted   Issuers:</span>
{verificationStats.totalIssuers}
              </div>
            </div>
          </div>
        )}

        {/* Add Trusted Issuer */}
        <div className="p-4 bg-gray-50 rounded-lg">
          <h5 className="font-medium text-gray-700 mb-2">Add Trusted Issuer</h5>
          <div className="space-y-2">
            <input
              type="text"
              placeholder="Issuer Name"
              value={newIssuer.name}
              onChange={(e) => setNewIssuer({ ...newIssuer, name: e.target.value })}
              className="w-full p-2 border border-gray-300 rounded-md"
            />
```

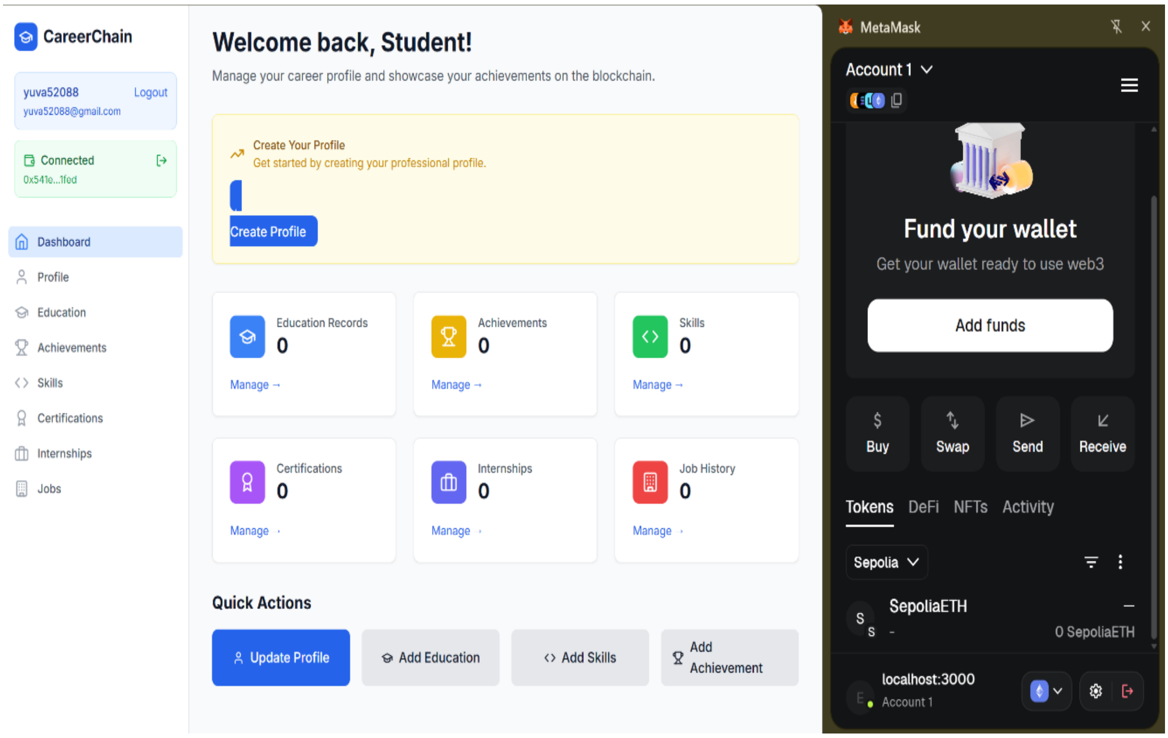
APPENDIX-2

LIST OF PUBLICATIONS

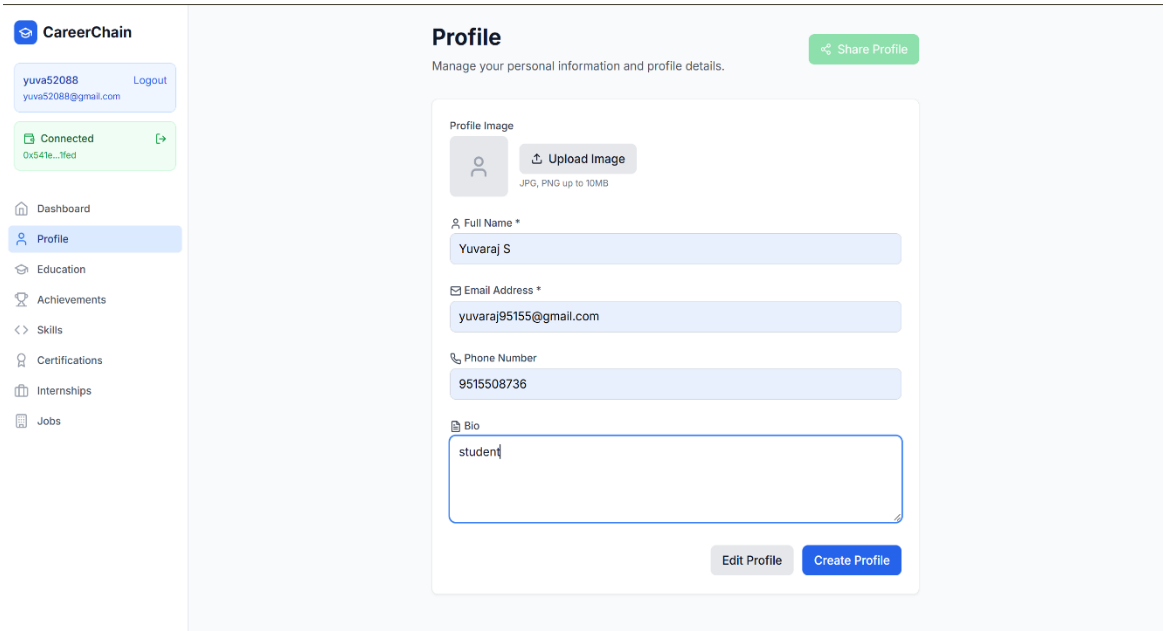
1. Karthikeyan S, Harshavarthan P, Saravanan V, Varshitha G, Varshika K, Vignesh T, Yuvaraj S, Manikkam V, “SMART TRAVEL COMPANION SYSTEM WITH BUDGET OPTIMIZATION USING RECOMMENDATIONS”, 7th International Conference on Artificial Intelligence and Smart Environments (ICAISE’25), held at Hammamet, Tunisia from November 6 to 8, 2025.

2. Sarathkumar R, MuthuPandian M, Roshan Kumar P, Sanjai D, Yabesh Martin P, Yokeshwaran K, Gokulavanan M, “AI-BASED VIRTUAL INTERVIEWER SYSTEM”,7th International Conference on Artificial Intelligence and Smart Environments (ICAISE’25), held at Hammamet, Tunisia from November 6 to 8, 2025.

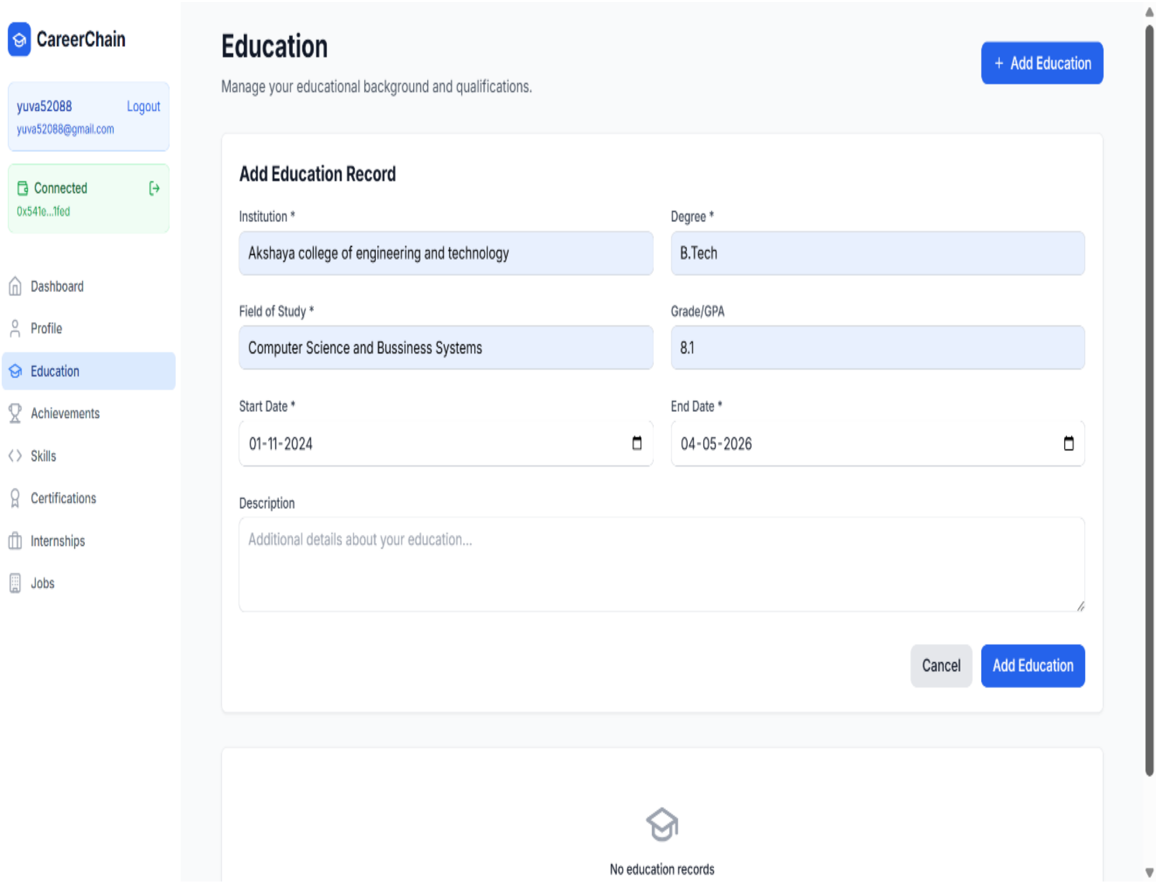
Web Dashboard



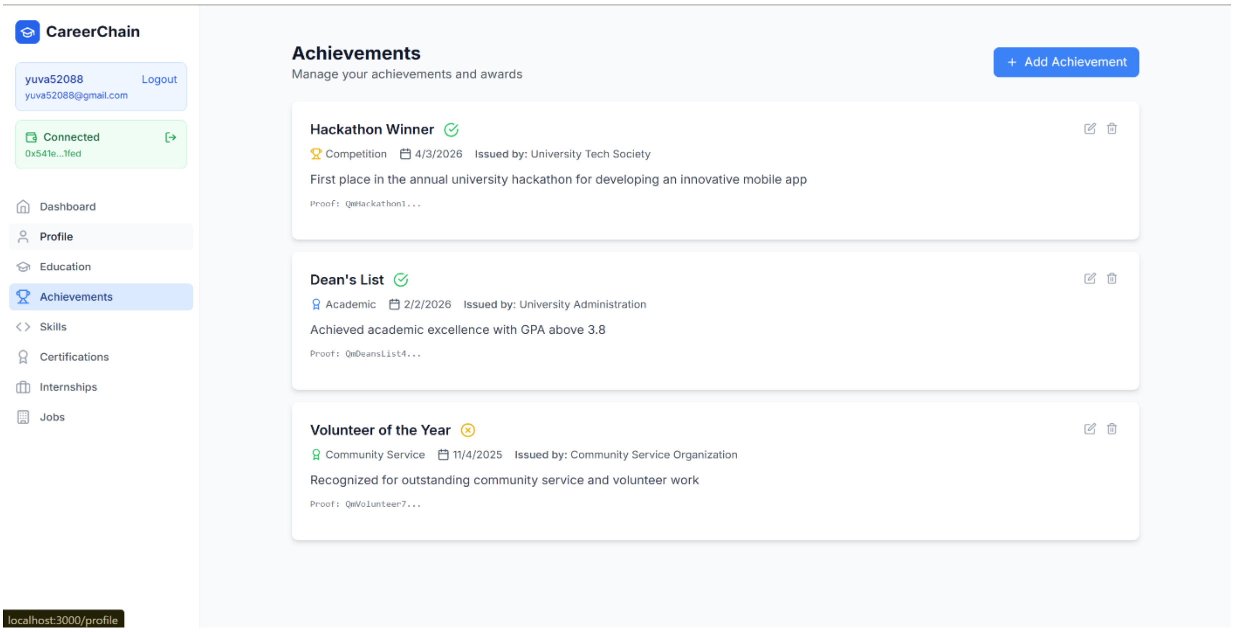
Profile page



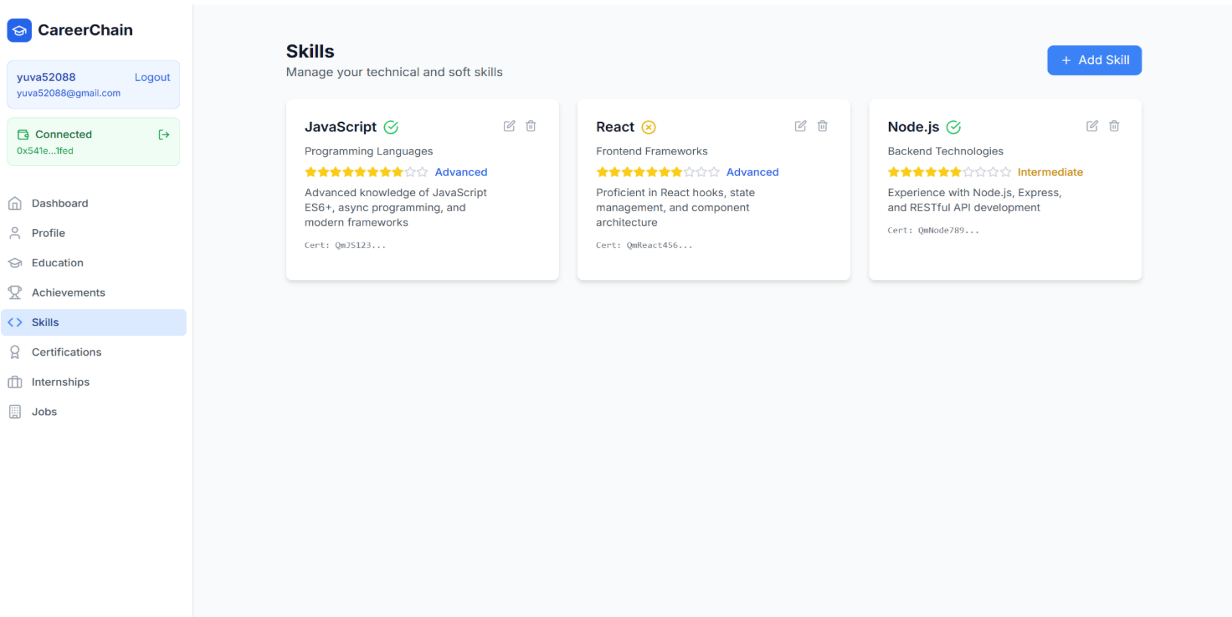
Education page



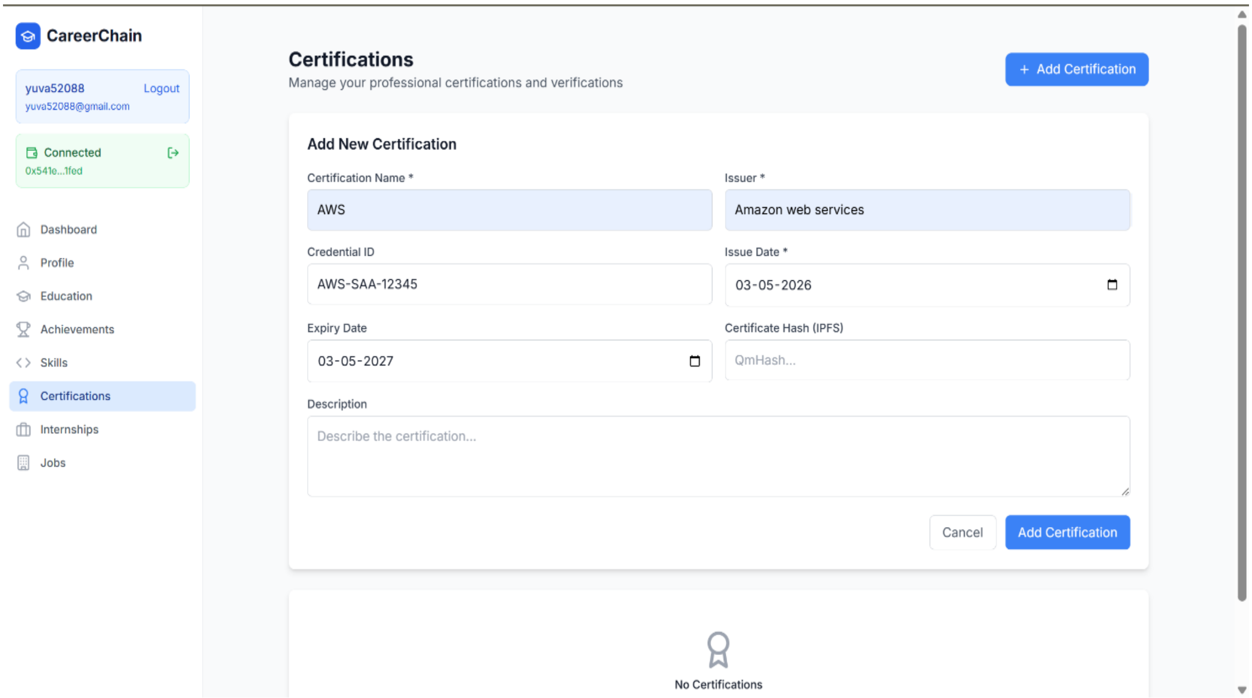
Achievement page



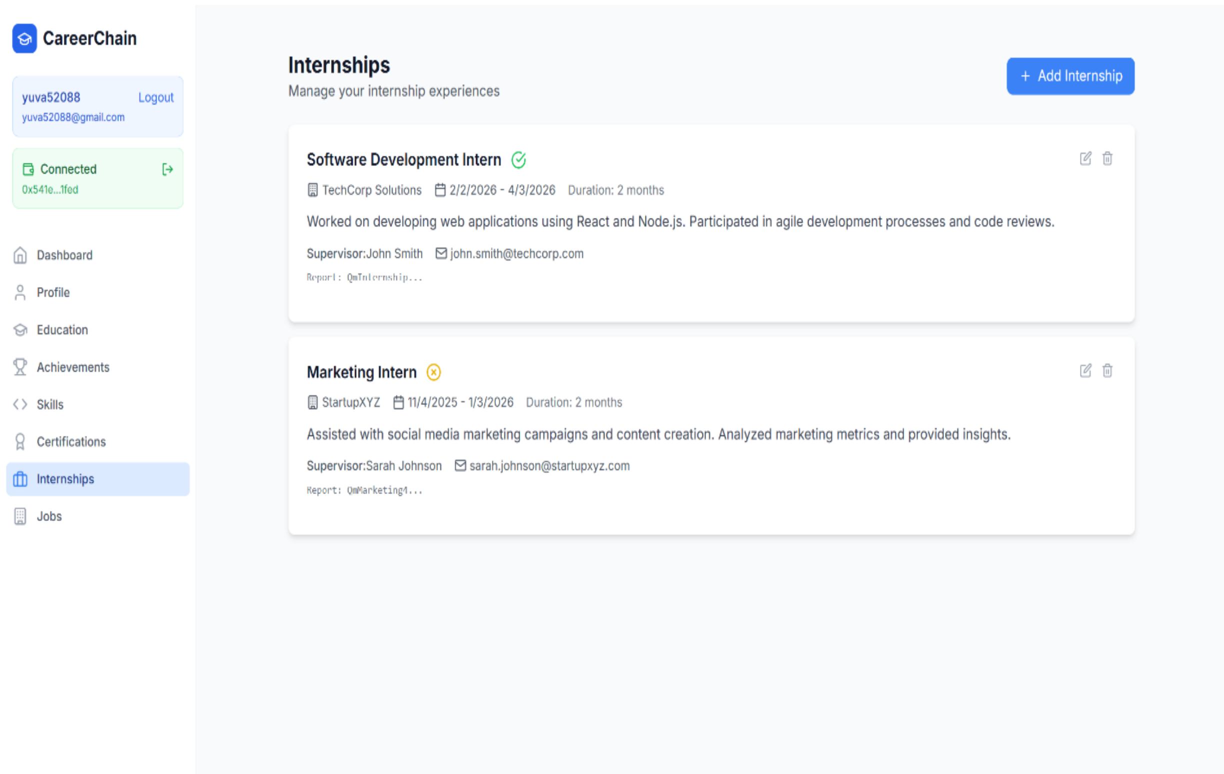
Skill page

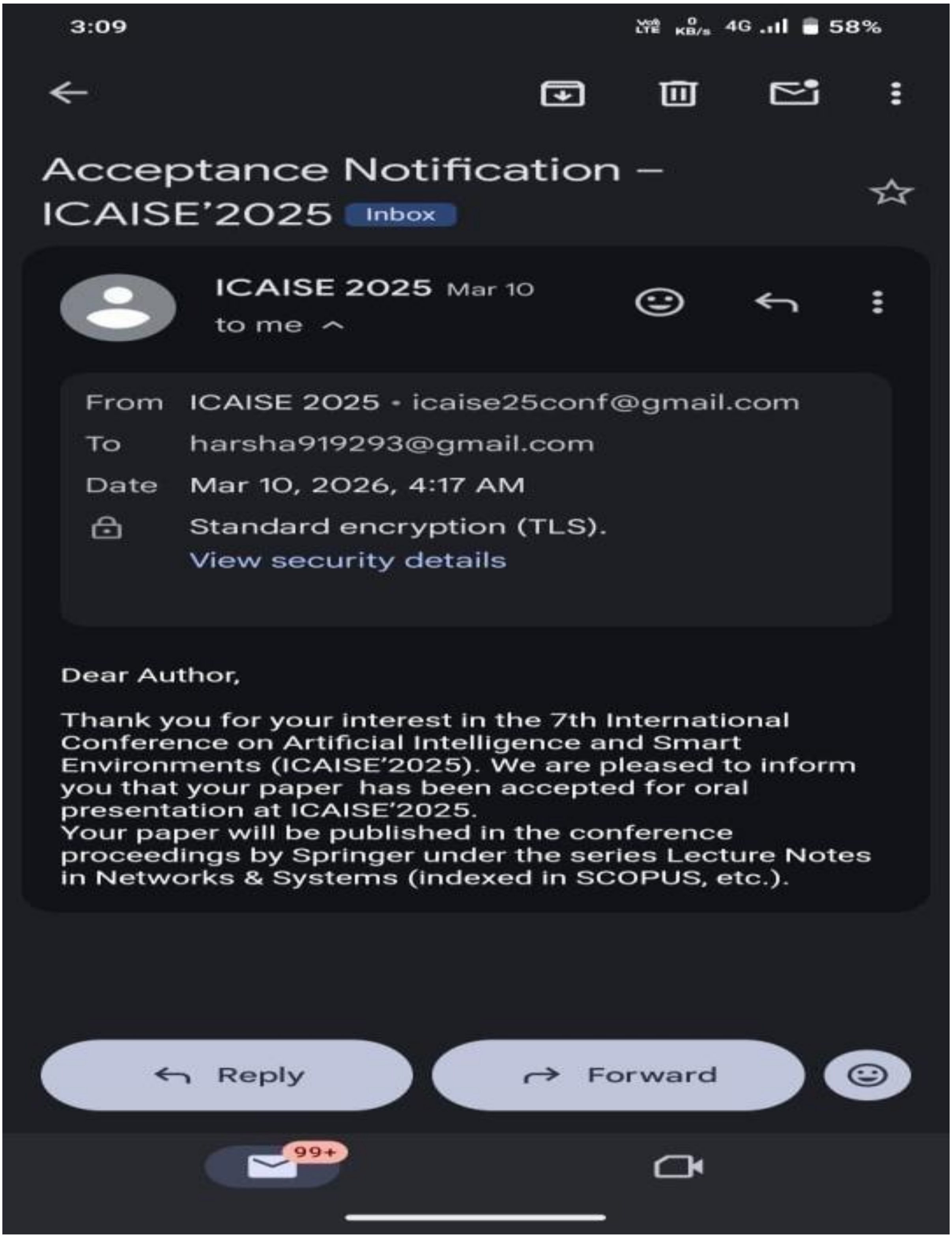


Certification page



Internship page





3:09 58%

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Dear Author,

Thank you for your interest in the 7th International Conference on Artificial Intelligence and Smart Environments (ICAISE'2025). We are pleased to inform you that your paper has been accepted for oral presentation at ICAISE'2025.

Your paper will be published in the conference proceedings by Springer under the series Lecture Notes in Networks & Systems (indexed in SCOPUS, etc.).

Reply Forward

99+

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" AI-Based Virtual Interviewer System"

Co-authors: **Malini Muthupandian, Roshan Kumar P, Sanjai D, Yabesh Martin P, Yokeshwaran K, Sarathkumar R**

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" Smart Travel Companion System with Budget Op-timization Using Recommendations "

Co-authors: **Harshavarthan P, Saravanan V, Varshitha G, Varshita K, Vignesh T, Yuvaraj S, Manikkam V**

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